Muting Durably Reduces Engagement with Untrustworthy Online Sources; Media Literacy Doesn't

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Exposure to untrustworthy information has been linked to opinion manipulation, such as susceptibility to fake news [1], reduced trust and confidence in elections [2, 3], and vaccine hesitancy [4, 5]. It is also associated with negative behavioral outcomes, such as increased sharing of untrustworthy news and greater use of toxic language [6], purchasing products with negative health implications [7, 8], and reduced COVID vaccination rates [9] as well as additional infections and deaths [10].

Exposure and consumption of untrustworthy content is highly concentrated among a small subset of users [11–14] who likely drive much of the downstream harm. Because users who engage with untrustworthy content play a role in propagating such content, they represent the population of greatest interest for mitigation interventions. Yet, users with low-quality information diets remain underrepresented in research samples as most studies have targeted general populations of users. A couple of exceptions involve interventions targeting users with high exposure to untrustworthy sources [14] or who followed accounts known to spread misinformation [15]. Despite growing concerns and mounting evidence linking exposure to untrustworthy information to negative outcomes, we know surprisingly little about how to most effectively decrease exposure to such information, particularly among susceptible media users.

We therefore target vulnerable users who engage with untrustworthy sources, testing the long-term effects of two interventions: a media literacy campaign that seeks to improve how users evaluate the information they encounter, and a muting treatment that alters the information environment by reducing exposure to problematic sources. Specifically, we conducted a preregistered large-scale field experiment on X (formerly Twitter) with 1,210 active adult users in the United States who had previously engaged with at least one untrustworthy information source. They were randomly assigned to one of three conditions: a muting intervention, a media literacy intervention, or a control condition. Participants in the muting condition were given incentives to keep a random subset of untrustworthy sources muted during the four-week treatment period. Participants in the media literacy condition received a series of tips designed to improve discernment of misleading content, while those in the control and muting conditions received placebo tips.

Our primary estimands are the effects of assignment to the muting or media literacy treatment on a range of behavioral and attitudinal outcomes. We estimate these effects using ordinary least squares (OLS) regression with heteroskedasticity-robust standard errors, controlling for a vector of pre-treatment covariates selected via lasso [16] for each outcome. We adjust for multiple testing using the sharpened false discovery rate procedure of Benjamini *et al.* [17].

Muting reduced the fraction of participants who engaged with any targeted sources six months

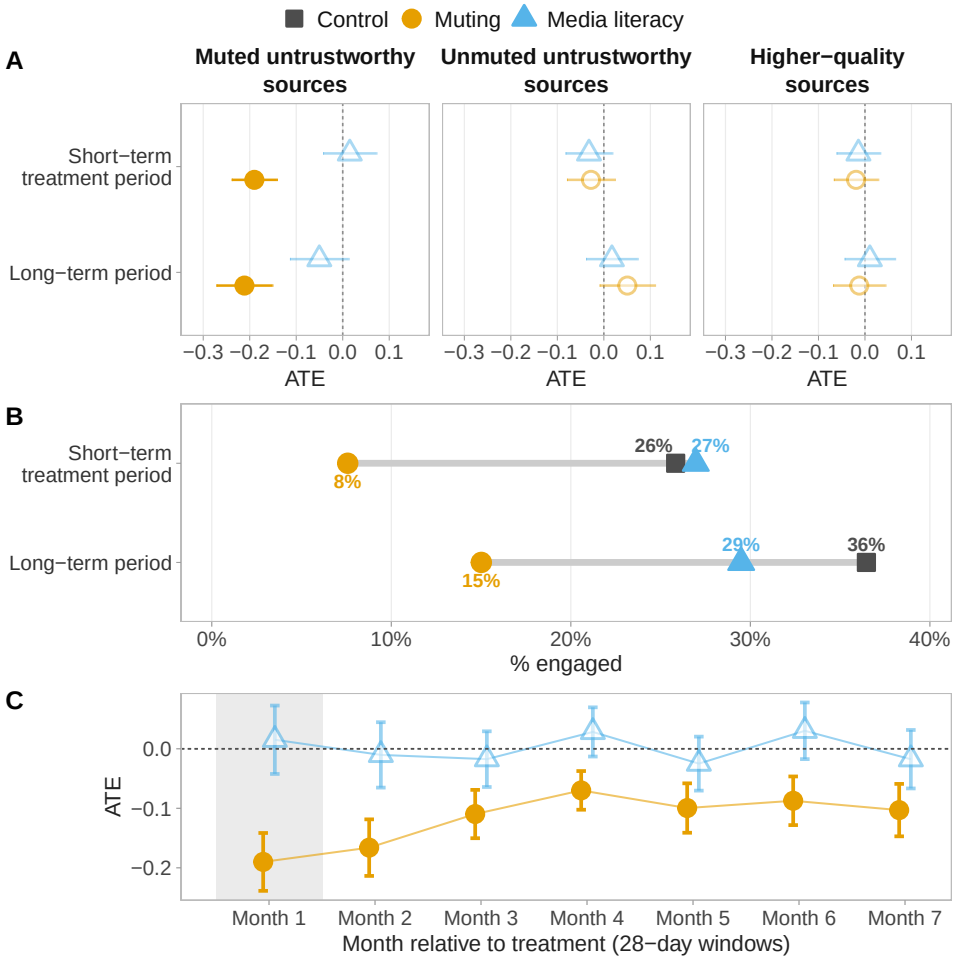


Figure 1: Treatment effects of muting and media literacy interventions on engagement behaviors in two time periods: the incentivized treatment period (four weeks) and the subsequent long-term period (six months). (A) Average treatment effects (ATEs) on engagement with muted untrustworthy sources, unmuted untrustworthy sources, and higher-quality sources relative to the control. ATE is defined in terms of the fraction of participants who had any engagement with these different kinds of sources: positive values indicate that the fraction increased in the treatment group. (B) Fractions of participants who engage with muted untrustworthy sources in the three conditions. (C) Exploratory monthly breakdown of ATEs of muting and media literacy interventions on engagement with muted sources relative to the control. The shaded area indicates the incentivized treatment period. In panels (A) and (C), error bars indicate 95% confidence intervals and filled symbols indicate significant effects ($p < 0.05$ after false discovery rate correction [17]).

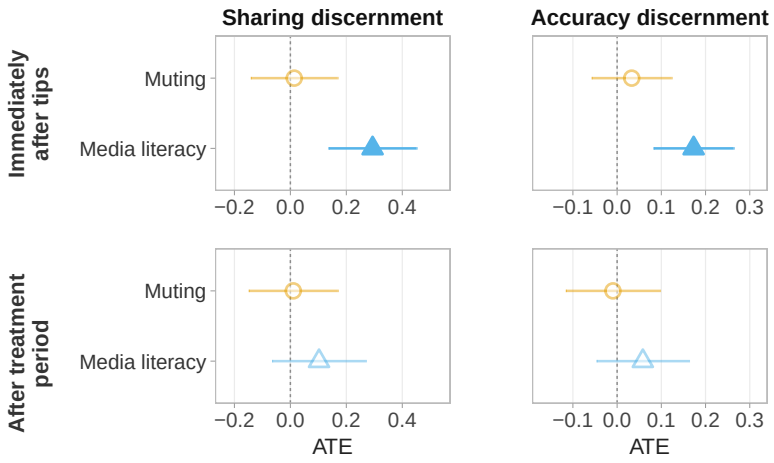


Figure 2: Treatment effects of muting and media literacy interventions on sharing and accuracy discernment across two time points: immediately after tips were shown and after the three weekly reminder tips during the treatment period. ATE is defined in terms of the level of sharing discernment (left column) and accuracy discernment (right column): positive values indicate that the treatment group has higher discernment. Error bars indicate 95% confidence intervals and filled symbols indicate significant effects ($p < 0.05$ after false discovery rate correction [17]).

beyond the incentive period. Approximately 15% of participants in the muting condition did so compared to 36% in the control condition — a relative reduction of approximately 58% (Figure 1B). The effect was largest during the incentive period (-0.19 ATE, $p < 0.001$) and narrowed before stabilizing in months 5–7 ($p < 0.001$), approximately half the size of the initial effect (Figure 1C). Although participants could unmute the targeted accounts at any time and incentives to keep them muted ended after four weeks, fewer than 11% of treated participants unmuted any of the targeted sources even six months after the incentive period ended.

We find no evidence that participants compensated for reduced exposure to the muted sources by shifting their engagement to other sources. Neither the muting nor the media literacy treatment induced substitution to other, unmuted untrustworthy sources (Figure 1A, middle column) or higher-quality news sources (last column).

The media literacy treatment did not affect downstream engagement behavior. Although it improved accuracy and sharing discernment immediately after participants received the tips (Figure 2), these effects did not persist after four weeks, despite reinforcement through reminders.

Our results suggest that consumption of untrustworthy information can be understood as environment-dependent behavior, driven by visibility and accessibility to such information. This visibility-driven information consumption mirrors findings in public health domains, where choice architecture is reported to have an impact on habitual consumption [18–20]. For example, structural interventions that limit access to junk food yield greater impact on dietary habits than literacy campaigns focused on identification or alternative-seeking [21]. A similar logic may apply to online information diets, where asking users to seek better sources requires greater effort, especially for those already embedded in untrustworthy media environments. The muting intervention may be durable precisely because it demands little effort. Media literacy, by contrast, demands sustained

effort to identify and avoid problematic content in real time, a cognitive burden that may explain its limited behavioral effects despite the discernment gains we observed immediately after treatment.

The practical implication of this study is that users can meaningfully customize their information environments through relatively low-effort tools such as muting. Importantly, our findings do not hinge on who implements these tools. The intervention may be offered directly by platforms, provided through third-party or civil society initiatives, or be adopted voluntarily by users themselves. Our argument also does not depend on adjudicating which sources should be muted. In past years, social media platforms had operationalized such definitions through moderation practices, including shadow banning and algorithmic downranking. Many platforms have moved away from such moderation efforts by curbing or eliminating fact-checking programs, cutting trust and safety staff, and rewarding low-quality content that maximizes engagement (including AI slop) via monetization. Instead, platforms could make it practical for users to curate their own information environment.

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